Arul Rhik Mazumder

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Education

Carnegie Mellon University **Aug. 2023 – May 2026**

Bachelor of Science in Computer Science, Dean's List High Honors *Pittsburgh, Pennsylvania*

Select Coursework: Probabilistic Graphical Models, Quantum Computing, Complexity Theory, Parallel Architecture and Algorithm Design, Deep Learning Systems

Activities: President of CMU Quantum, Quant Club, ACM @ CMU, Project Olympus

Experience

Caltech SURF **May 2025 - August 2025**

Undergraduate Research Fellow *Pasadena, California*

- Worked under Samson Wang with John Preskill's group to design early fault-tolerant quantum primitives based on Trotterization achieving exponential improvement in circuit depth compare to other modern techniques (pdf, slides).
- Participated in a quantum algorithms reading group focused on QSVT, QSP, QET, and other related methods.

Duke Software-Tailored Architectures for Quantum Codesign (STAQ) **June 2025**

NSF-Funded Invited Scholar *Durham, North Carolina*

- Completed a PhD-level, week-long quantum computing seminar featuring advanced lectures on quantum simulation, error correction, and architecture, taught by leading researchers including Peter Love, Kenneth Brown, and Yu Tong.
- Visited the Duke Quantum Center to learn about experimental research on diverse hardware platforms, including ion traps and neutral atoms. Applied what I learned to write (pdf, pdf, pdf, slides, slides, slides) for CMU Quantum.

International Business Machines (IBM) **May 2024 – August 2024**

Research Intern *Cambridge, Massachusetts*

- Researched a novel learning algorithm for discrete graphical models using samples generated from Glauber Dynamics.
- Implemented end-to-end sampling, learning, and evaluation pipelines in a dedicated research codebase (GitHub).

Carnegie Mellon University Quantum Technologies Group **May 2023 – Current**

Research Intern *Pittsburgh, Pennsylvania*

- Produced (tutorial) for quantum solutions to optimization problems on IBM and D-Wave machines (GitHub).
- To be (presented) at the 2025 INFORMs Annual Meeting and published in 2025 TutORials in Operations Research.

Select Projects/Publications

Quantum Annealing for solving the Shipment Rerouting | *Python, D-Wave Ocean, MILP* **January 2025**

- Developed both classical approximation algorithms and quantum annealing algorithms for the shipment rerouting problem using D-Wave's quantum annealer and custom mixed-integer linear programming models (paper).
- Demonstrated quantum solutions achieving near-optimal performance faster than classical solvers in various scenarios.

Benchmarking Metaheuristic-Integrated QAOA against Quantum Annealing | *Python, Deep Learning* **May 2024**

- Numerically benchmarked the performance of metaheuristic optimizers for optimizing QAOA circuits (paper).
- Published in Intelligent Computing Proceedings of the 2024 Computing Conference, Volume 3.
- Accepted to Quantum Information Processing (QIP) 2024 as Poster Presentation.

A Hybrid Quantum Algorithm for the Metric Traveling Salesman Problem | *Python, Ocean* **January 2023**

- Researched and developed a novel quantum-classical hybrid algorithm for the Traveling Salesman Problem. Numerical tests on D-Wave backends for small input sizes demonstrate 3× improvement (paper).
- Published in 2023 IEEE International Parallel and Distributed Processing Symposium (IPDPS).

Comparisons of Classic and Quantum String Matching Algorithms | *Python, Qiskit* **August 2022**

- Developed novel quantum string-matching algorithm using Qiskit and tested it on IBM Quantum Backends (paper).
- Published in proceedings of ACM's 4th International Conference on Advanced Information Science and System (AISS).

Awards

Olympiads: USA(J)MO (2021-22), USAPhO Bronze (2022), USACO Gold (2023), USNCO National Finalist (2022)

Other Competitions: 2nd Place overall at YQuantum (code) - The Yale Quantum Institute Grand Prize Recipient (2025),
2nd Place at Correlation One and Citadel Summer Invitational Datathon (report) (2024),
Graduate Student Association Best Campus Infrastructure Hack Award at Hack@CMU (report) (2023),
Regeneron Science Talent Search Scholar [Top 300] (2023), Purple Comet 5th Place (2022),
International Math Modelling Competition National Finalist (2022),
United States Math Talent Search Honorable Mention and Bronze (2021-22)